# DATA ANALYTICS

**PECST523**

# 1. The Data Analytics Lifecycle

A useful way to hold the six phases in your head is to follow one running example through all of them. Suppose a telecom company wants to know why customers are leaving ("churning") for a rival network.

## Phase 1: Discovery

The team meets business stakeholders, learns the domain (billing cycles, network coverage, customer support logs), takes stock of what resources exist — people, computing power, existing data, and time — and turns the vague worry "customers are leaving" into a testable statement, for instance: "Customers who call support more than three times in a month are more likely to churn within 90 days." This testable statement is called an Initial Hypothesis (IH). The team also identifies which data sources might exist: call detail records, billing history, complaint tickets, competitor pricing.

## Phase 2: Data Preparation

A separate analytic sandbox is set up so raw production data is never touched directly. Data is loaded largely as it is and only then transformed inside the sandbox — this order gives the process its name, ETLT (Extract, Transform, Load, Transform), as opposed to the older ETL order used in traditional data warehousing. Inside the sandbox the team performs data conditioning: removing duplicate customer records, fixing inconsistent date formats, handling missing call-duration values, and flagging obvious outliers (a call that supposedly lasted 40 hours). This phase overlaps heavily with exploratory data analysis (EDA) — plotting histograms of call frequency, checking value ranges, and looking for anything that looks wrong before any model is built.

## Phase 3: Model Planning

The team studies how variables relate to one another — does complaint frequency correlate with churn? does contract length matter more than price? — and chooses which variables to keep and which modelling approach (logistic regression for a yes/no churn outcome, a decision tree, or a clustering approach to first segment customers) best fits the business question and the data available. Tools commonly named here include R, SQL, and SAS.

## Phase 4: Model Building

The prepared data is split into a training set and a test set. The model chosen in Phase 3 is actually fitted on the training data, then checked against the test data. If a laptop cannot handle the volume of call-detail records, the team may need a more powerful, possibly parallel, computing environment at this stage.

## Phase 5: Communicate Results

The team sits down again with the stakeholders from Phase 1 and asks, in plain business language, whether the model is actually useful: does it identify churn-risk customers early enough for a retention offer to matter? Key findings are quantified in money terms ("this model could save an estimated 2 crore rupees a year in retained subscription revenue"), and every important assumption, caveat, and limitation of the model is written down honestly.

## Phase 6: Operationalize

Final code, reports, and technical documentation are handed over, and often a pilot run is carried out on a small slice of real customers before the churn-prediction model is switched on for the whole customer base. Running a contained pilot first surfaces practical problems — slow scoring, a dashboard nobody actually reads — while the cost of a mistake is still small.

## The Feedback Loops (a point examiners specifically look for)

The six phases are usually drawn as a ring, not a straight line, because real projects step backward constantly:

- Model Planning back to Data Preparation — while choosing variables, the team discovers that call-duration data has too many missing values to be useful, and has to return to Phase 2 to re-clean or re-source it.
- Model Building back to Model Planning — while fitting the model, the team finds that a decision tree overfits badly and returns to Phase 3 to try logistic regression instead, or to drop a noisy variable.
- Communicate Results back to any earlier phase — if the churn model turns out to explain the data well but is useless for the business (say, it only flags customers after they have already left), the team can loop all the way back to Discovery and reframe the question.

***Sketching this for the essay question:*** *draw six boxes in a circle (Discovery, Data Preparation, Model Planning, Model Building, Communicate Results, Operationalize), connect them with the forward arrows going around the ring, and then add two short backward arrows: Model Planning → Data Preparation, and Model Building → Model Planning. Label each backward arrow in one phrase with why the loop happens.*

## Neighbouring lifecycle models

Two related process models are worth knowing by name, since examiners sometimes ask for a comparison:

- CRISP-DM (Cross-Industry Standard Process for Data Mining): Business Understanding, Data Understanding, Data Preparation, Modeling, Evaluation, Deployment — conceptually very close to the six-phase lifecycle above, just with slightly different phase boundaries.
- KDD (Knowledge Discovery in Databases): Selection, Preprocessing, Transformation, Data Mining, Interpretation/Evaluation — an older, narrower process that treats "data mining" as one step inside a larger pipeline, rather than the whole pipeline.

# 2. Data Types, Attributes, and Summary Statistics

Every distance formula, every chart type, and every statistical test in this guide is valid only for certain attribute types. Getting this classification right is usually worth marks on its own, and it is the first thing to check before answering any computational question.

## The Four Attribute Types

| Type | Definition | Everyday example | Operations that make sense |
|---|---|---|---|
| Nominal | Named categories, no order | Blood group, district of residence, SIM card operator | Equality only (count, mode) |
| Ordinal | Ordered categories, unequal gaps | Movie rating (1–5 stars), exam grade (A/B/C), pain scale | Order comparisons, median, rank correlation |
| Interval | Ordered, equal gaps, no true zero | Temperature in Celsius, year on a calendar | Addition, subtraction, mean |
| Ratio | Ordered, equal gaps, true zero | Height, salary, number of calls made, distance travelled | All arithmetic, ratios meaningful (“twice as tall”) |

A quick test for interval versus ratio: does “zero” mean a genuine absence of the quantity? 0°C does not mean “no temperature” (it is an arbitrary point, which is why “20°C is not twice as hot as 10°C”), whereas 0 rupees in a bank balance does mean an absence of money, so money is a ratio attribute.

## Discrete, Continuous, and Binary Attributes

– Continuous: can take any real value in a range, such as a person's height (165.3 cm) or the time taken to load a web page (0.482 seconds).

– Discrete: takes a finite or countable set of values, such as the number of siblings a person has, or the number of times a web page times out in a day.

– Binary: a discrete attribute with exactly two states, coded 0 and 1. Binary attributes are further split into:

  - Symmetric — both states carry equal weight, e.g. “Gender” coded Male = 1, Female = 0, or a coin toss Heads/Tails. Neither state is more “important” than the other.
  - Asymmetric — one state is rare or more informative than the other, e.g. a COVID test result (Positive = 1 is rare and far more informative than Negative = 0), or “item bought in this transaction” in a supermarket basket (a 1 tells you something interesting; the huge number of 0s, for items not bought, tells you almost nothing).

## Measures of Central Tendency and Dispersion (closely related, frequently a viva topic)

Once an attribute's type is known, the appropriate “typical value” and “spread” statistics follow directly.

| Statistic | Formula / definition | Appropriate for |
|---|---|---|
| Mean | $\mu = (\Sigma x_i) / n$ | Interval, ratio |
| Median | middle value when data is sorted (average of two middle values if n is even) | Ordinal, interval, ratio |

| Statistic | Formula / definition | Appropriate for |
|---|---|---|
| Mode | most frequently occurring value | Nominal, ordinal, interval, ratio |
| Range | maximum − minimum | Interval, ratio |
| Variance / Std. deviation | $\sigma^2 = \Sigma(x_i-\mu)^2/n$ ; $\sigma = \sqrt{\sigma^2}$ | Interval, ratio |
| Interquartile range (IQR) | Q3 − Q1 (spread of the middle 50% of data) | Ordinal, interval, ratio |

**Worked example: mean, median, mode, range**

Daily number of support calls received over 9 days: 12, 15, 12, 18, 20, 12, 25, 14, 16.

Sorted: 12, 12, 12, 14, 15, 16, 18, 20, 25.

- Mean = (12+12+12+14+15+16+18+20+25)/9 = 144/9 = 16
- Median = the middle (5th) value in the sorted list = 15
- Mode = 12 (appears three times, more than any other value)
- Range = 25 − 12 = 13

Real-life reading: the mean (16 calls/day) is pulled upward by the unusually busy day of 25 calls, while the median (15) is not affected by that one extreme day — which is exactly why news reports usually quote “median household income” rather than “mean household income”: a handful of very wealthy households would otherwise distort the mean.

# 3. Distance and Similarity Measures for Continuous Data

Distance measures turn two rows of numeric data into a single number that says how different the two rows are. They are the basic building block behind nearest-neighbour classification, clustering, and recommendation systems.

## The Minkowski Family

$$d(P, Q) = \left( \Sigma_i |p_i - q_i|^h \right)^{(1/h)}, \quad \text{for } P = (p_1,\dots,p_n) \text{ and } Q = (q_1,\dots,q_n)$$

Manhattan, Euclidean, and Chebyshev distance are simply this one formula evaluated at three different values of h.

| Name | h | Formula | Everyday picture |
|---|---|---|---|
| Manhattan ($L_1$, “city-block”) | 1 | $\Sigma_i \lvert p_i - q_i \rvert$ | Distance walked between two points in a city where you can only move along streets laid out in a grid — you cannot cut diagonally through a building. |
| Euclidean ($L_2$) | 2 | $\sqrt{\Sigma_i (p_i - q_i)^2}$ | Straight-line distance — how far apart two points are “as the crow flies,” or the distance a drone would fly directly between two rooftops. |
| Chebyshev ($L\infty$) | $\to \infty$ | $\max_i \lvert p_i - q_i \rvert$ | The number of moves a king needs on a chessboard — it can move diagonally, so only the larger of the row-difference and column-difference matters. |

**Worked example 1**

$P_1 = (3, 2, 6, 1)$, $P_2 = (7, 5, 2, 4)$. Coordinate-wise absolute differences: $|3-7|=4$, $|2-5|=3$, $|6-2|=4$, $|1-4|=3$.

- Manhattan = 4+3+4+3 = 14
- Euclidean = $\sqrt{(4^2+3^2+4^2+3^2)} = \sqrt{(16+9+16+9)} = \sqrt{50} \approx 7.07$
- Chebyshev = max(4,3,4,3) = 4

**Worked example 2 (a second, differently-shaped, data set for practice)**

A = (10, 0, 4), B = (2, 8, 0), representing three product-rating scores (out of 10) given by two customers. Differences: $|10-2|=8$, $|0-8|=8$, $|4-0|=4$.

- Manhattan = 8+8+4 = 20
- Euclidean = $\sqrt{(64+64+16)} = \sqrt{144} = 12$
- Chebyshev = max(8,8,4) = 8

Real-life reading: food-delivery apps use something close to Manhattan distance to estimate delivery time in a gridded city layout, mapping apps use Euclidean distance as the “as the crow flies” baseline before road distance is calculated, and Chebyshev distance appears literally in chess engines and in warehouse-robot path planning where diagonal movement is allowed and costs the same as a straight move.

## Cosine Similarity (closely related, common follow-up question)

Unlike the Minkowski distances above, cosine similarity ignores the magnitude of the two vectors and looks only at the angle between them — useful when two documents should be considered "similar" even if one is much longer than the other, as long as they use words in similar proportions.

$$\cos(\theta) = (P \cdot Q) / (\|P\| \times \|Q\|)$$

**Worked example**

P = (3, 0, 4), Q = (1, 0, 2) (word-frequency counts for two short documents, over three key terms).

$P \cdot Q = 3\times1 + 0\times0 + 4\times2 = 3+0+8 = 11$. $\|P\| = \sqrt{(9+0+16)} = 5$. $\|Q\| = \sqrt{(1+0+4)} = \sqrt{5} \approx 2.236$.

$$\cos(\theta) = 11 / (5 \times 2.236) = 11 / 11.18 \approx 0.984$$

A value close to 1 means the two documents point in almost the same direction in word-frequency space, i.e. they are topically very similar, even though document P uses each term roughly three times more often than document Q. Search engines, plagiarism checkers, and "recommended for you" engines on shopping and streaming sites all lean on cosine similarity for exactly this reason.

***Also worth knowing:*** *all Minkowski distances obey the four metric axioms — non-negativity, d(P,P) = 0, symmetry (d(P,Q) = d(Q,P)), and the triangle inequality. For a mix of numeric and categorical columns in the same table, Gower's distance is the standard generalisation, and for very high-dimensional sparse data (like text), cosine similarity is usually preferred over Euclidean distance because Euclidean distance becomes unreliable as dimensions increase — a phenomenon often called the curse of dimensionality.*

# 4. Similarity Measures for Binary Attributes

For two binary vectors of equal length, first tally how the two vectors agree and disagree, position by position, using a 2×2 contingency table.

| | **y = 1** | **y = 0** | **Row total** |
|---|---|---|---|
| x = 1 | q (both 1) | r (x=1, y=0) | q + r |
| x = 0 | s (x=0, y=1) | t (both 0) | s + t |
| Column total | q + s | r + t | n = q+r+s+t |

*Simple Matching Coefficient* $SMC = (q + t) / (q + r + s + t)$

*Jaccard Coefficient* $J = q / (q + r + s)$ *Jaccard Distance* $= 1 - J = (r + s) / (q + r + s)$

SMC counts a shared "0" as just as meaningful as a shared "1," so it belongs with symmetric binary attributes. Jaccard ignores t (both-zero) entirely, which is why it belongs with asymmetric binary attributes, where a shared absence tells you nothing interesting.

**Worked example 1**

x = (1,1,0,1,0,1,0), y = (1,0,1,1,0,0,1). Position-by-position comparison:

| **Position** | **1** | **2** | **3** | **4** | **5** | **6** | **7** |
|---|---|---|---|---|---|---|---|
| x | 1 | 1 | 0 | 1 | 0 | 1 | 0 |
| y | 1 | 0 | 1 | 1 | 0 | 0 | 1 |
| Category | q | r | s | q | t | r | s |

q = 2, r = 2, s = 2, t = 1 (check: 2+2+2+1 = 7, matches the vector length).

- SMC = (2+1)/7 = 3/7 ≈ 0.4286
- Jaccard J = 2/(2+2+2) = 2/6 = 1/3 ≈ 0.333
- Jaccard Distance = 1 − 1/3 = 2/3 ≈ 0.667

**Worked example 2 (asymmetric, real-world flavour)**

Two supermarket baskets recorded over the same five products (Milk, Bread, Eggs, Rice, Soap), 1 = purchased: Basket A = (1,1,0,0,1), Basket B = (1,0,0,1,1).

Position-wise: Milk both 1 (q), Bread A=1,B=0 (r), Eggs both 0 (t), Rice A=0,B=1 (s), Soap both 1 (q). So q=2, r=1, s=1, t=1.

- SMC = (2+1)/5 = 3/5 = 0.60
- Jaccard J = 2/(2+1+1) = 2/4 = 0.50

Real-life reading: because "both customers did NOT buy soap-flavoured cereal" (a shared zero among thousands of possible grocery items) says nothing about the two customers' tastes, supermarket "customers who bought this also bought" engines use Jaccard-style similarity, not SMC, on purchase-basket data. SMC is more at home comparing, say, two patients' yes/no answers to a symmetric checklist such as "do you own a car (yes/no)" and "do you own a bicycle (yes/no)", where a shared "no" is just as informative as a shared "yes."

***Also worth knowing:*** *the Dice (Sørensen) coefficient, $2q/(2q+r+s)$, is a close cousin of Jaccard that weights matches more heavily and is common in image-segmentation accuracy scoring; DNA and protein sequence comparison in bioinformatics, and spell-checker/autocorrect suggestion ranking, both rely on binary or set-overlap similarity measures of this family.*

# 5. Discrete Random Variables and Probability Distributions

A discrete random variable X lists every possible outcome together with the probability of that outcome (its probability mass function, or PMF). A valid PMF must have every probability between 0 and 1, and all probabilities must add up to exactly 1.

$$E(X) = \mu = \Sigma_i\, x_i \cdot P(x_i) \qquad Var(X) = \sigma^2 = \Sigma_i\, x_i^2\, P(x_i) - [E(X)]^2 \qquad \sigma = \sqrt{Var(X)}$$

**Worked example 1**

X = number of network interruptions in a day, with P(X) = {0.50, 0.30, 0.15, 0.05} for X = {0,1,2,3}.

| $x_i$ | $P(x_i)$ | $x_i \cdot P(x_i)$ | $x_i^2 \cdot P(x_i)$ |
|---|---|---|---|
| 0 | 0.50 | 0.00 | 0.00 |
| 1 | 0.30 | 0.30 | 0.30 |
| 2 | 0.15 | 0.30 | 0.60 |
| 3 | 0.05 | 0.15 | 0.45 |
| Total | 1.00 | 0.75 | 1.35 |

- E(X) = 0.75 interruptions per day
- Var(X) = $1.35 - 0.75^2 = 1.35 - 0.5625 = 0.7875$, so $\sigma \approx 0.887$

**Worked example 2 (an insurance-style payout, a very common real-world use of E(X))**

An insurer sells a policy where, in a given year, a customer files 0 claims with probability 0.7, 1 claim (payout ₹10,000) with probability 0.2, and 2 claims (payout ₹20,000) with probability 0.1.

Let X = payout amount, so X = {0, 10000, 20000} with P(X) = {0.7, 0.2, 0.1}.

- E(X) = $0\times0.7 + 10000\times0.2 + 20000\times0.1 = 0 + 2000 + 2000$ = ₹4,000

Real-life reading: this expected payout of ₹4,000 per customer per year is exactly the kind of number an insurance company uses as the starting point for setting the annual premium — they must charge more than ₹4,000 on average simply to break even, before adding a margin for profit and administrative cost. The same E(X) idea drives expected-value thinking in lotteries, casino games, stock-market expected returns, and A/B-testing decisions in product design.

## Two Named Distributions Worth Knowing (closely related)

The distribution in the worked examples above was given directly as a table. In practice, discrete data very often follows one of two standard named patterns:

- Binomial distribution: counts the number of "successes" in n independent yes/no trials, each with the same success probability p. Example: the number of defective items in a batch of 20, if each item independently has a 5% chance of being defective. Mean = np, Variance = np(1−p).
- Poisson distribution: counts the number of events happening in a fixed interval of time or space, when events happen independently at a constant average rate λ. Example: the number of network interruptions in an hour, or the number of customers arriving at a bank counter in ten minutes. Mean = Variance = λ, which is a useful giveaway sign that a data set might be Poisson-distributed.

# 6. The Normal Distribution and Hypothesis Testing

The Normal (Gaussian) distribution is the familiar symmetric bell curve, fully described by its mean $\mu$ and standard deviation $\sigma$. Standardising any Normal variable with $z = (x-\mu)/\sigma$ converts it to the Standard Normal distribution ($\mu = 0$, $\sigma = 1$), for which critical values are tabulated.

## Vocabulary

| Term | Meaning |
|---|---|
| Null hypothesis, $H_0$ | the default claim being tested, e.g. "this coin is fair" or "the new drug has no effect" |
| Alternative hypothesis, $H_1$ | what is concluded if $H_0$ is rejected |
| Significance level, $\alpha$ | the risk of a Type I error the tester is willing to accept, chosen before the test, commonly 0.05 or 0.01 |
| Critical (rejection) region | the tail area(s) of the distribution where, if the test statistic falls there, $H_0$ is rejected |
| Acceptance region | the central area; landing here means $H_0$ is not rejected |
| Type I error | rejecting $H_0$ when $H_0$ is actually true — probability = $\alpha$ |
| Type II error | failing to reject $H_0$ when $H_0$ is actually false — probability = $\beta$ |

## Sketching the Two-Tailed Diagram (exam requirement)

Draw one symmetric bell curve. Mark the centre as $\mu$ (or 0, if standardised). Mark two critical values, $-z\alpha/2$ and $+z\alpha/2$, symmetric about the centre. Shade the two small tail areas beyond these critical values and label them "reject $H_0$" — together these two tails have area $\alpha$, split as $\alpha/2$ in each tail because the test is two-tailed. Leave the large central area unshaded (or differently shaded) and label it "acceptance region — fail to reject $H_0$." Type II error, $\beta$, is normally shown by drawing a second, shifted bell curve (representing the true, alternative distribution) and shading the part of that second curve that falls inside the first curve's acceptance region — that overlap is exactly the probability of wrongly failing to reject $H_0$.

**<u>Worked example: a two-tailed z-test</u>**

A machine is supposed to fill packets with $\mu = 500$ g of flour, with a known population standard deviation $\sigma = 10$ g. A sample of $n = 25$ packets has a sample mean of 495 g. Test at $\alpha = 0.05$ whether the machine is mis-calibrated ($H_0$: $\mu = 500$; $H_1$: $\mu \neq 500$).

$$z = (\bar{x} - \mu) / (\sigma/\sqrt{n}) = (495 - 500) / (10/\sqrt{25}) = -5 / 2 = -2.5$$

For a two-tailed test at $\alpha = 0.05$, the critical values are $\pm1.96$. Since $-2.5$ falls beyond $-1.96$ (it is further into the rejection tail), the sample mean falls in the critical region, so $H_0$ is rejected: there is evidence at the 5% level that the machine is mis-calibrated.

Real-life reading: this exact procedure — sample, compute z (or t), compare to a critical value — is what quality-control teams on a factory floor, and pharmaceutical labs testing whether a batch of tablets has the correct active-ingredient content, run every day.

## Common Two-Tailed Critical Values

| α | Critical z-value (two-tailed) |
|---|---|
| 0.10 | ±1.645 |
| 0.05 | ±1.96 |
| 0.01 | ±2.576 |

***Also worth knowing:*** *in a one-tailed test the entire α sits in a single tail, so the one-tailed critical value at a given α is always closer to the mean than the corresponding two-tailed critical value. The p-value is the probability, assuming $H_0$ is true, of seeing a result at least as extreme as the one observed; $H_0$ is rejected whenever the p-value is smaller than α. Lowering α to reduce Type I error, without increasing the sample size, always raises the Type II error rate β — the two cannot both be shrunk for free.*

# 7. Contingency Tables and Joint / Marginal Distributions

A contingency table cross-tabulates two categorical variables, recording how many observations fall into every combination of categories. It is the basic tool behind every joint-probability, conditional-probability, and association question in Module 2.

**Worked example 1**

200 users, split by age group and channel preference: 110 Young shoppers (80 Online, 30 In-Store) and 90 Adult shoppers (30 Online, 60 In-Store).

| Age Group | Online | In-Store | Row Total |
|---|---|---|---|
| Young | 80 | 30 | 110 |
| Adult | 30 | 60 | 90 |
| Column Total | 110 | 90 | 200 |

- Joint frequency — the count in an inner cell, e.g. 80 users are Young AND Online.
- Marginal frequency — the row or column totals, e.g. 110 users are Young regardless of channel; 110 users prefer Online regardless of age.

Dividing every cell by the grand total (200) converts this into the joint probability table:

| Age Group | P(Online) | P(In-Store) | P(row) |
|---|---|---|---|
| Young | 0.40 | 0.15 | 0.55 |
| Adult | 0.15 | 0.30 | 0.45 |
| P(column) | 0.55 | 0.45 | 1.00 |

**Worked example 2 (a classic public-health style table, for extra practice)**

300 adults surveyed on smoking status and history of a chest infection: of 90 smokers, 54 had a chest infection and 36 did not; of 210 non-smokers, 42 had a chest infection and 168 did not.

| Smoking status | Infection | No infection | Row Total |
|---|---|---|---|
| Smoker | 54 | 36 | 90 |
| Non-smoker | 42 | 168 | 210 |
| Column Total | 96 | 204 | 300 |

Joint probability P(Smoker, Infection) = 54/300 = 0.18. Marginal probability P(Smoker) = 90/300 = 0.30, P(Infection) = 96/300 = 0.32.

# 8. Conditional Probability, Independence, and the Chi-Square Test

$$P(A \mid B) = P(A \text{ and } B) / P(B)$$

In words: restrict attention to the cases where B happened, then ask what fraction of those cases also had A.

**Worked example 1 (continuing the shoppers table)**

$$P(\text{Online} \mid \text{Young}) = P(\text{Young, Online}) / P(\text{Young}) = 0.40 / 0.55 \approx 0.727$$

$$P(\text{In-Store} \mid \text{Young}) = 0.15 / 0.55 \approx 0.273 \qquad (0.727 + 0.273 = 1, \text{ as it must})$$

This is identical to working with raw counts: 80/110 = 0.727 and 30/110 = 0.273 — conditioning on "Young" simply means dividing by the Young row's own total, not by the grand total.

**Worked example 2 (a second, three-category conditional frequency table)**

150 employees, Job Satisfaction (Low/Medium/High) by Experience Level: Junior — 20 Low, 40 Medium, 20 High (row total 80); Senior — 10 Low, 20 Medium, 40 High (row total 70).

| Experience | Low | Medium | High | Row Total |
|---|---|---|---|---|
| Junior | 20 | 40 | 20 | 80 |
| Senior | 10 | 20 | 40 | 70 |
| Column Total | 30 | 60 | 60 | 150 |

Conditional frequency distribution (dividing each row by its own total):

| Experience | P(Low \| row) | P(Medium \| row) | P(High \| row) |
|---|---|---|---|
| Junior (÷80) | 25.0% | 50.0% | 25.0% |
| Senior (÷70) | 14.3% | 28.6% | 57.1% |

As experience rises from Junior to Senior, the probability of High satisfaction more than doubles (25.0% to 57.1%) while Low satisfaction nearly halves (25.0% to 14.3%). Because both variables are ordinal and the shift is consistently in one direction, this is a positive ordinal association — stated and justified with these actual percentages, not just "by eye."

## Independence and the Chi-Square Test of Independence

Two categorical variables are statistically independent only if P(A | B) equals P(A) for every category — that is, knowing B tells you nothing about A. In both worked tables above the conditional distributions clearly differ from the overall marginal distribution, so the variables are not independent. The chi-square test gives a formal, numerical way to check this rather than relying on inspection.

$$\text{Expected count } E(i,j) = (\text{Row Total}_i \times \text{Column Total}_j) / \text{Grand Total} \qquad \chi^2 = \Sigma\, (O - E)^2 / E$$

**Worked example: chi-square, using the smoking / infection table**

Expected count for Smoker & Infection, under the assumption of independence: E = (90×96)/300 = 28.8. Similarly E(Smoker, No infection) = (90×204)/300 = 61.2; E(Non-smoker, Infection) = (210×96)/300 = 67.2; E(Non-smoker, No infection) = (210×204)/300 = 142.8.

| Cell | Observed (O) | Expected (E) | $(O-E)^2/E$ |
|---|---|---|---|
| Smoker, Infection | 54 | 28.8 | 22.05 |

| **Cell** | **Observed (O)** | **Expected (E)** | **(O−E)²/E** |
|---|---|---|---|
| Smoker, No infection | 36 | 61.2 | 10.37 |
| Non-smoker, Infection | 42 | 67.2 | 9.45 |
| Non-smoker, No infection | 168 | 142.8 | 4.45 |

$$\chi^2 = 22.05 + 10.37 + 9.45 + 4.45 \approx 46.3$$

With 1 degree of freedom (a 2×2 table has df = (rows−1)(columns−1) = 1), the critical $\chi^2$ value at $\alpha = 0.05$ is 3.84. Since 46.3 is far larger than 3.84, the null hypothesis of independence is rejected: smoking status and chest infection are strongly associated in this data — the same style of test that underlies most published medical-study headlines linking a habit or exposure to a health outcome.

# 9. Data Visualisation: Bar Charts and Neighbouring Chart Types

Bar charts display categorical (nominal or ordinal) data. The exam asks specifically about two close variants.

## Grouped (Clustered) Bar Chart

For each category on the x-axis, draw two or more bars side by side, one bar per sub-group, all measured against a shared y-axis. Because every bar shares the same baseline, grouped bar charts are the easiest way to compare the exact size of one sub-group against another, both within a category and across categories.

## 100% Stacked Bar Chart

For each category on the x-axis, draw a single bar, stretched to a fixed total height of 100%, and split into coloured segments whose heights represent the percentage share of each sub-group. This trades away the ability to read exact totals (every bar has the same height) in exchange for making the composition, the relative proportion of each sub-group, very easy to compare from one category to the next.

**Worked example: customer-feedback dataset**

Product Category A: Satisfied = 60, Dissatisfied = 40 (total 100). Product Category B: Satisfied = 30, Dissatisfied = 70 (total 100).

– Grouped bar chart: two bars for Category A (heights 60 and 40) beside two bars for Category B (heights 30 and 70), all read off a 0–100 count axis.
– 100% stacked bar chart: Category A is a single bar split into a 60% segment (Satisfied) and a 40% segment (Dissatisfied); Category B is a single bar split into a 30% segment and a 70% segment; both bars reach exactly 100%.

When sketching either chart: label the x-axis with the two product categories, label the y-axis (“number of customers” for the grouped chart, “percentage of customers” for the stacked chart), add a legend mapping shading or hatching to Satisfied/Dissatisfied since this is a black-and-white sketch, and write the actual value or percentage on or beside each bar or segment.

## Structural Differences

| Aspect | Grouped bar chart | Stacked bar chart |
|---|---|---|
| Bars per category | several, side by side | one, divided into segments |
| Reading a sub-group's exact value | easy for every sub-group, since all bars share the axis as baseline | easy only for the bottom segment; higher segments must be read as a difference between two boundaries |
| Reading the category total | not shown directly | shown directly as the full bar height (or 100% if normalised) |
| Best used when | the absolute size of each sub-group is what matters, and there are only a few sub-groups | the overall composition, or how a proportion changes across categories, is what matters most |

## Neighbouring Chart Types (closely related)

- Pie chart — shows part-to-whole composition for a single category only; a 100% stacked bar chart is generally preferred over several pie charts when the same composition needs to be compared across more than one category.
- Histogram — for continuous numeric data grouped into bins; unlike a bar chart, the bars touch, because the x-axis is a continuous scale, not separate categories. Example: the distribution of exam marks across a class.
- Box plot (box-and-whisker plot) — summarises a continuous variable's median, quartiles, and outliers in one compact shape; commonly drawn side by side across categories to compare spread, e.g. comparing salary spread across different job roles.
- Scatter plot — plots two continuous variables against each other to visually check for a relationship, e.g. advertising spend versus monthly sales; this is the usual first step before computing a correlation coefficient.

# 10. Measuring Association Between Two Variables

This section widens the conditional-probability idea from Section 8 into the fuller toolkit used whenever a question asks you to say whether, and how strongly, two variables are related.

## For two categorical variables

- Chi-square test of independence (Section 8) — the general-purpose numerical test for any two categorical variables, nominal or ordinal.
- Odds ratio — for a 2×2 table specifically, using the q, r, s, t labels from Section 4: odds ratio = (q×t)/(r×s). A value of 1 means no association; well above 1 means the two '1' outcomes tend to occur together more than chance would predict.

**Worked example: odds ratio, smoking / infection table**

$$\text{Odds ratio} = (54 \times 168) / (36 \times 42) = 9072 / 1512 = 6.0$$

A smoker in this sample has six times the odds of having had a chest infection compared with a non-smoker — the same style of number reported in epidemiological studies linking a risk factor to a health outcome.

## For two ordinal variables

- Conditional frequency comparison (Section 8) — the simplest approach: check whether the conditional probability of the "high" outcome rises or falls consistently as the row variable moves through its ordered levels.
- Spearman's rank correlation and Kendall's tau — correlation computed on ranks rather than raw values, so they capture any consistent (monotonic) trend, not only a straight-line one.
- Gamma statistic — based on counting concordant versus discordant pairs of observations, ranging from −1 to +1 like a correlation coefficient.

## For two continuous variables: Pearson Correlation Coefficient

$$r = \Sigma(x_i-\bar{x})(y_i-\bar{y}) / \sqrt{[\,\Sigma(x_i-\bar{x})^2 \times \Sigma(y_i-\bar{y})^2\,]}$$

**Worked example**

Advertising spend (₹ thousand) and monthly sales (₹ lakh) over five months: (2,10), (4,15), (6,17), (8,22), (10,26).

$\bar{x} = 6$, $\bar{y} = 18$. Deviations from the mean and their products:

| x | y | $x-\bar{x}$ | $y-\bar{y}$ | product | $(x-\bar{x})^2$ |
|---|---|---|---|---|---|
| 2 | 10 | −4 | −8 | 32 | 16 |
| 4 | 15 | −2 | −3 | 6 | 4 |
| 6 | 17 | 0 | −1 | 0 | 0 |
| 8 | 22 | 2 | 4 | 8 | 4 |
| 10 | 26 | 4 | 8 | 32 | 16 |

Sum of products = 32+6+0+8+32 = 78. Sum of $(x-\bar{x})^2$ = 16+4+0+4+16 = 40. Sum of $(y-\bar{y})^2$ = 64+9+1+16+64 = 154.

$$r = 78 / \sqrt{(40 \times 154)} = 78 / \sqrt{6160} = 78 / 78.5 \approx 0.994$$

An r this close to +1 says advertising spend and sales move together almost perfectly in this small sample — the same statistic a marketing analyst would quote to justify (or question) an advertising budget increase.

# 11. Formula Sheet

## Distances (continuous data)

| Measure | Formula |
|---|---|
| Manhattan ($L_1$) | $\Sigma\lvert p_i - q_i\rvert$ |
| Euclidean ($L_2$) | $\sqrt{\Sigma(p_i - q_i)^2}$ |
| Chebyshev ($L\infty$) | $\max\lvert p_i - q_i\rvert$ |
| Cosine similarity | $(P\cdot Q)/(\lVert P\rVert\lVert Q\rVert)$ |

## Binary similarity

| Measure | Formula |
|---|---|
| SMC | (q+t)/(q+r+s+t) |
| Jaccard coefficient | q/(q+r+s) |
| Jaccard distance | (r+s)/(q+r+s) |
| Dice coefficient | 2q/(2q+r+s) |

## Probability and statistics

| Measure | Formula |
|---|---|
| Expected value | $E(X) = \Sigma\, x_i P(x_i)$ |
| Variance | $Var(X) = \Sigma\, x_i^2 P(x_i) - [E(X)]^2$ |
| z-score | $z = (x-\mu)/\sigma$ |
| z-test statistic | $z = (\bar{x}-\mu)/(\sigma/\sqrt{n})$ |
| Conditional probability | $P(A\mid B) = P(A\cap B)/P(B)$ |
| Expected count ($\chi^2$) | E(i,j) = Row Total × Col Total / Grand Total |
| Chi-square statistic | $\chi^2 = \Sigma(O-E)^2/E$ |
| Odds ratio (2×2 table) | (q×t)/(r×s) |
| Pearson correlation | $r = \Sigma(x-\bar{x})(y-\bar{y}) / \sqrt{[\Sigma(x-\bar{x})^2\Sigma(y-\bar{y})^2]}$ |

## Hypothesis testing

| Term | Key fact |
|---|---|
| Type I error | probability $\alpha$; rejecting a true $H_0$ |
| Type II error | probability $\beta$; failing to reject a false $H_0$ |
| Two-tailed critical z | $\alpha$=0.10 → ±1.645; $\alpha$=0.05 → ±1.96; $\alpha$=0.01 → ±2.576 |

## Data Analytics Lifecycle, in order

Discovery → Data Preparation → Model Planning → Model Building → Communicate Results → Operationalize, with feedback loops Model Planning → Data Preparation and Model Building → Model Planning.